D07 · Order Management System (OMS/ATP/CTP)

03 · Order prioritization, allocation & backorder management

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Order prioritization and allocation decide which orders receive inventory when supply is short, using explicit rules — customer segment, contract, urgency, penalties — and manage backorders with reliable availability dates.

2 WHY IT MATTERS

Without rules, scarce inventory goes first-come, first-served, which rarely matches business priorities. **Allocation under scarcity is a commercial decision and should be approved in S&OP before the shortage, not improvised during it.** For backorders, the credibility of the promised date matters as much as the delay itself.

3 HOW IT IS DONE

1. **Segment customers and orders.** Strategic value, contractual obligations, urgency and penalty exposure.
2. **Approve allocation rules in S&OP.** Documented priority order, partial-shipment rules and exceptions.
3. **Configure the rules in the OMS.** Automatic allocation and backorder handling, with manual override logged.
4. **Communicate backorders proactively.** Projected dates from the real supply plan, updated as it changes.

IN PRACTICE

Define OMS prioritization rules with commercial and SC teams before scarcity occurs

Prioritization rules under scarcity are business decisions — not operational decisions. They must be approved in the S&OP as company policy before a real scarcity situation requires applying them under pressure.

Communicate backorders proactively to the customer — don’t wait for the customer to ask

Proactive backorder communication (with projected availability date and reason) converts bad news into a customer service gesture. Reactive communication converts a delay into a complaint.

Backorder management requires the OMS to be integrated with the supply plan

The projected availability date for the backorder must be calculated with real production or replenishment plan data — not a generic “2–3 week” estimate. An OMS integrated with the APS can calculate the real availability date.

Analyze the backorder root cause — it’s a symptom of an upstream problem

A high backorder rate is a symptom of: insufficient safety stock, incorrect forecast, or supply constraint. The corrective action is in the inventory policy or the supply plan — not in the OMS.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Backorder management in scarcity — pharmaceutical manufacturer, 3 weeks without production due to equipment failure — Manufacturer has 180 active orders and only 40% of the inventory needed to cover them.

Prioritization rule applied	Orders affected	OMS decision
Priority 1: Hospitals and pharmacies with contractual SLA (penalty if not fulfilled)	42 orders · 100% of available inventory assigned first to this segment	42 orders fulfilled 100% · $0 contractual penalty
Priority 2: A-distributors (>80% of revenue)	38 orders · Remaining inventory assigned by order size	28 orders covered 100% · 10 orders in partial backorder (60–80%)

Prioritization rule applied	Orders affected	OMS decision
Priority 3: B and C distributors	100 orders · No inventory available	100 orders on backorder · Projected availability date: 18 days

Result: With OMS prioritization: $0 in contractual penalties. The 28 key A-distributors received available inventory. B and C distributors received proactive communication of the projected availability date 3 weeks in advance — minimizing damage to commercial relationships.

HOW IT IS MEASURED

Backorder Rate % (orders that cannot be served immediately)

Backorder Rate %(Orders entering backorder / Total orders for the period) × 100

Benchmark: <3% for operations with well-managed inventory · <8% in planned demand peaks A sustained backorder rate >10% indicates inventory is not aligned with demand — from insufficient safety stock, incorrect forecast, or a structural supply constraint.

Backorder Fill Rate % (backorders fulfilled on the projected date)

Backorder Fill Rate %(Backorders fulfilled on the projected availability date communicated to the customer / Total backorders for the period) × 100

Benchmark: >90% of backorders fulfilled on the projected date · A backorder date that also isn't met is a double failure The credibility of backorder management depends on the projected date communicated to the customer being accurate. A backorder whose date is updated 3 times generates more dissatisfaction than the original backorder.

WHAT YOU WOULD USE

Backorder Management

SAP S/4HANA — Backorder Processing (BOP) — Module: Automated Backorder Management + Priority RulesSAP S/4HANA BOP module automatically manages backorders with configured prioritization rules, reallocates available inventory by priority, and updates projected availability dates.

Manhattan Active Omni — Order Promising — Module: Backorder + Promise ManagementManhattan system with backorder and delivery promise management integrating current inventory, in-transit, and the production plan.

4 SO WHAT

Decide priorities before the shortage. Segment, approve the rules in S&OP, encode them in the OMS and give backordered customers dates they can trust.

From the SCRA v2.2 reference architecture · mrsupplychain.ai · © 2026 Ismael López · Mr. Supply Chain® México